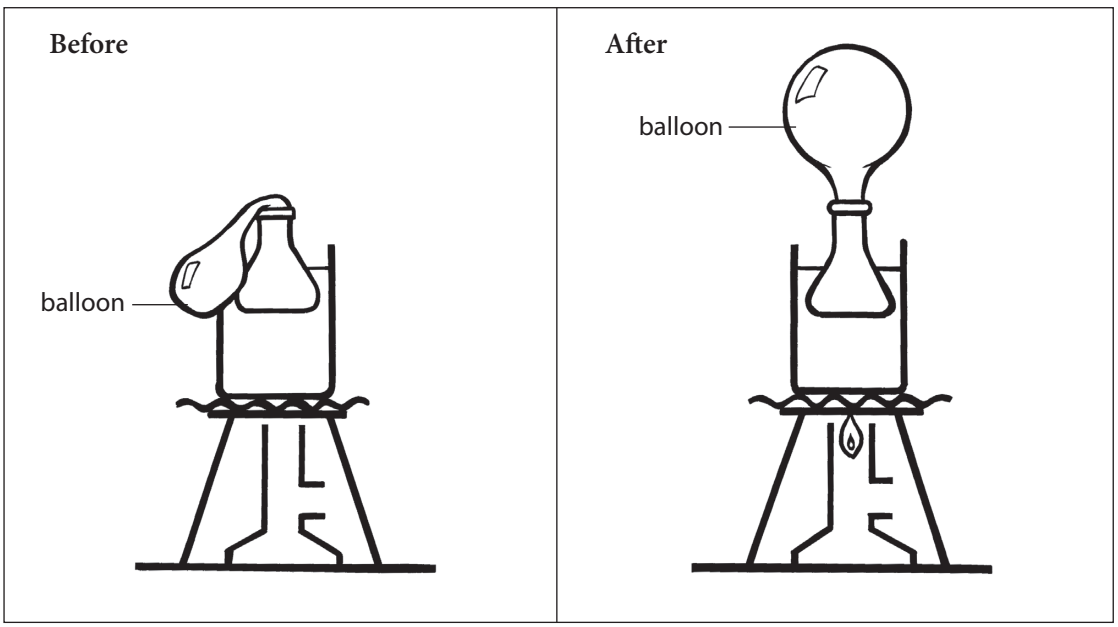
Watch me expand

When a solid, liquid or gas is heated, its particles vibrate or move more rapidly and push each other further apart.

1 The diagrams below show three separate experiments in which a gas, a liquid and a solid are heated. Use the diagrams to help you complete the procedure for each experiment. You need to write the aim, the steps involved in setting up the experiment, the results that you observed and then a conclusion for each experiment.

Experiment 1



Aim – what are you trying to do?

Equipment – what things do you need?

Method – what are the steps involved?

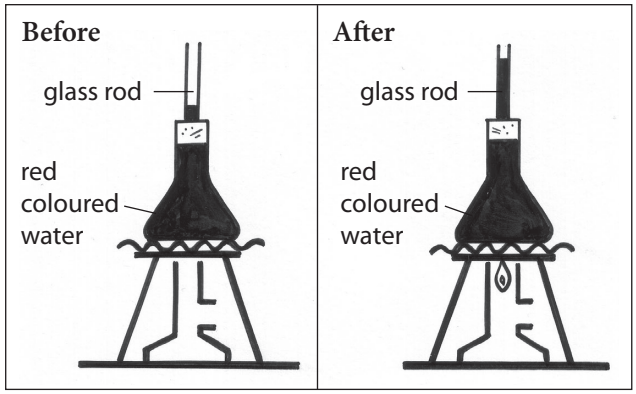
Results – what did you observe?

Conclusion – what have you learnt?

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Experiment 2



Aim: _____

_____

Equipment: _____

_____

_____

Method: _____

_____

_____

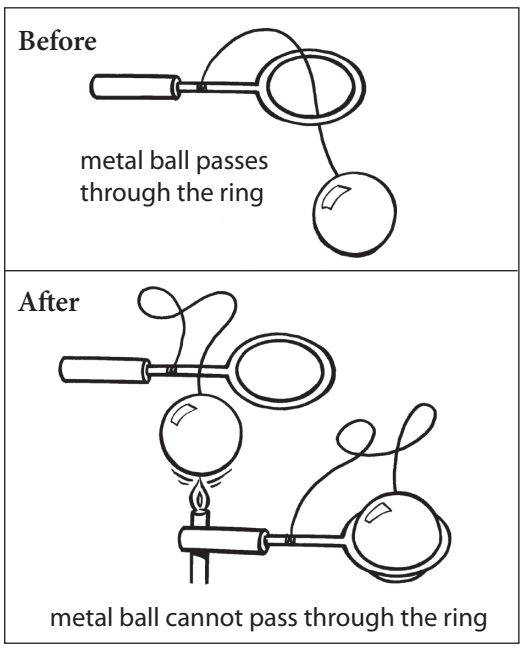
Results: _____

_____

Conclusion: _____

_____

Experiment 3



Aim: _____

_____

Equipment: _____

_____

_____

Method: _____

_____

_____

_____

_____

_____

Results: _____

_____

Conclusion: _____

_____

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2 **a** What does **contract** mean? Find out and write a definition.

_____

b The experiments clearly show that solids, liquids and gases expand. Can they be contracted?_____

c How could you use each experiment to show contraction? Record your ideas in the table.

How to show contraction		
Experiment 1	**Experiment 2**	**Experiment 3**

d Cross out the incorrect terms to make the statements true.

In order to make solids, liquids and gases expand, [heating / cooling] is required. This causes the particles to move [more slowly / faster]. To make solids, liquids and gases contract, [heating / cooling] is required. This causes the particles to move [more slowly / faster].

e From your observations, which state of matter expanded the most? Justify your answer.

_____

_____

People deal with expanding and contracting solids, liquids and gases every day and use their knowledge to design and build new things. For example, when new bridges are built, expansion gaps must be included, so the bridge does not buckle when it expands in hot weather. Thermometers contain a liquid, either mercury or coloured alcohol, in a bulb connected to a narrow tube. When the temperature rises, the level of the liquid rises as well. Another example is a hot air balloon – it is filled with gas and when the gas is heated, it expands. The expanded air becomes less dense than the colder air, making the balloon rise to great heights.

3 List at least three other examples where expansion and contraction are involved.

_____

_____

_____

_____

4 Describe at least two situations where you have noticed solids, liquids and gases around you behaving differently on a cold day or a warm day.

_____

_____

_____

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